EXP.NO: 9	Develop an Android Calculator Application using Android Studio
DATE:	

AIM:

To develop an Android application that performs basic arithmetic operations such as Addition, Subtraction, Multiplication, and Division using Android UI components like EditText, Button, and TextView.

ALGORITHM:

1. Start the application.
2. Open Android Studio and create a new project.
3. Design the UI using:
   o Two EditText fields for input o Buttons for operations (+, –, *, /)
   o One TextView to display result
4. Declare variables for UI components.
5. Get user input from EditText fields.
6. Convert input values into numeric format.
7. Set OnClickListener for each button.
8. Perform operations based on button clicked:
   o Addition → num1 + num2
   o Subtraction → num1 – num2
   o Multiplication → num1 × num2
   o Division → num1 / num2
9. Check division by zero.
10. Display result in TextView.
11. Provide Clear button to reset fields.
12. Stop the program.

CODE:

```
package com.example.webtech

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
```

```
import androidx.compose.foundation.layout.*
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Modifier
import androidx.compose.ui.unit.dp
import androidx.compose.ui.unit.sp

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContent {
            CalculatorApp()
        }
    }
}

@Composable
fun CalculatorApp() {
    var num1 by remember { mutableStateOf("") }
    var num2 by remember { mutableStateOf("") }
    var result by remember { mutableStateOf("Result will appear here") }

    Column(
        modifier = Modifier
            .fillMaxSize()
            .padding(20.dp),
        verticalArrangement = Arrangement.spacedBy(10.dp)
    ) {

        OutlinedTextField(
            value = num1,
            onValueChange = { num1 = it },
            label = { Text("Enter First Number") }
        )

        OutlinedTextField(
            value = num2,
            onValueChange = { num2 = it },
            label = { Text("Enter Second Number") }
        )
```

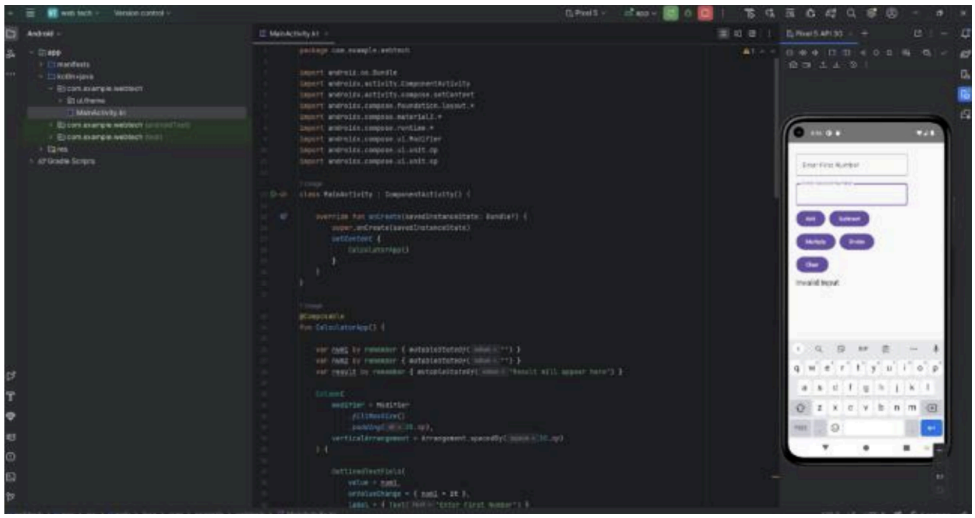
```
Row(horizontalArrangement = Arrangement.spacedBy(10.dp)) {

    Button(onClick = {
        calculate(num1, num2, "+") { result = it }
    }) {
        Text("Add")
    }

    Button(onClick = {
        calculate(num1, num2, "-") { result = it }
    }) {
        Text("Subtract")
    }
}

Row(horizontalArrangement = Arrangement.spacedBy(10.dp)) {

    Button(onClick = {
        calculate(num1, num2, "*") { result = it }
    }) {
        Text("Multiply")
    }

    Button(onClick = {
        calculate(num1, num2, "/") { result = it }
    }) {
        Text("Divide")
    }
}

Button(onClick = {
    num1 = ""
    num2 = ""
    result = "Result will appear here"
}) {
    Text("Clear")
}

Text(
```

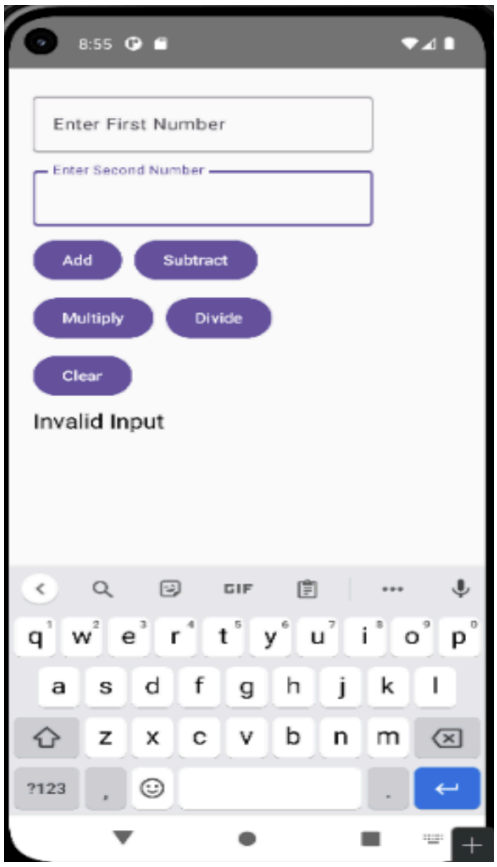
```
            text = result,
            fontSize = 20.sp
        )
    }
}
```

Calculation Function

```
fun calculate(n1: String, n2: String, op: String, update: (String) -> Unit) {

    if (n1.isEmpty() || n2.isEmpty()) {
        update("Please enter both numbers")
        return
    }

    try {
        val num1 = n1.toDouble()
        val num2 = n2.toDouble()

        val result = when (op) {
            "+" -> num1 + num2
            "-" -> num1 - num2
            "*" -> num1 * num2
            "/" -> {
                if (num2 == 0.0) {
                    update("Cannot divide by zero")
                    return
                }
                num1 / num2
            }
            else -> 0.0
        }

        update("Result: $result")

    } catch (e: Exception) {
        update("Invalid Input")
    }
}
```

OUTPUT:





RESULT:

The Android calculator application was successfully developed using EditText, Button, and TextView controls. The application performs basic arithmetic operations such as addition, subtraction, multiplication, and division, and displays accurate results.